

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<tool_call> Reasoning process here :

We are asked to find the de Broglie wavelength of an electron with a kinetic energy of1 eV.

The de Broglie wavelength is given by :

$\lambda = h / p$

where :

- h = Planck's constant = 6.626×10^{-34} J·s
- p = momentum of the particle

For a non-relativistic electron , kinetic energy $K = (1/2)mv^2$

We can express momentum $p = mv$, so :

$p = \sqrt{2mK}$

Thus , the de Broglie wavelength becomes :

$\lambda = h / \sqrt{2mK}$

We need to compute this with $K = 1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$

Mass of electron $m = 9.109 \times 10^{-31} \text{ kg}$

Now plug in the values :

$\lambda = (6.626 \times 10^{-34}) / \sqrt{2 \times 9.109 \times 10^{-31} \times 1.602 \times 10^{-19}}$

First compute the denominator :

$2 \times 9.109 \times 10^{-31} \times 1.602 \times 10^{-19} = 2 \times 9.109 \times 1.602 \times 10^{-50}$
 $\approx 2 \times 14.6 \times 10^{-50} = 29.2 \times 10^{-50} = 2.92 \times 10^{-49}$

Now $\sqrt{2.92 \times 10^{-49}} \approx \sqrt{2.92} \times 10^{-24.5} \approx 1.71 \times 10^{-24.5}$

Better to compute numerically :

$\sqrt{2.92 \times 10^{-49}} = \sqrt{2.92} \times 10^{-24.5} \approx 1.71 \times 10^{-24.5}$

But $10^{-24.5} = 10^{-24} \times 10^{-0.5} \approx 10^{-24} \times 0.316 = 3.16 \times 10^{-25}$

So $\sqrt{2.92 \times 10^{-49}} \approx 1.71 \times 3.16 \times 10^{-25} \approx 5.4 \times 10^{-25}$

Wait ⚡ better direct calculation :

$\sqrt{2.92 \times 10^{-49}} = \sqrt{2.92} \times 10^{-24.5}$

$\sqrt{2.92} \approx 1.71$

$10^{-24.5} = 10^{-(24+0.5)} = 10^{-24} \times 10^{-0.5} \approx 10^{-24} \times 0.316$
 $= 3.16 \times 10^{-25}$

So $\sqrt{2.92 \times 10^{-49}} \approx 1.71 \times 3.16 \times 10^{-25} \approx 5.4 \times 10^{-25}$

Now :

$\lambda = 6.626 \times 10^{-34} / (5.4 \times 10^{-25}) \approx 1.23 \times 10^{-9} \text{ m}$

Convert to cm : 1.23